Top 10 Genes Ranked by Potency of Perturbation (Sorted by BMD Median)

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Lung

Date: 22 May 2025
Time: 12:24:09 PM

Female

Gene Symbols	Entrez Gene IDs	Probe IDs	BMD_{rd25} ($BMDL_{rd25}$ - $BMDU_{rd25}$) in ppm	Maximum Fold Change	Direction of Expression Change
Ddit4	140942	DDIT4_8450	92.8 (13.6 - 256.6)	2.2	DOWN
Junb	24517	JUNB_8939	121.9 (39.9 - 278.5)	1.8	UP
Cebpd	25695	CEBPD_8283	132.6 (41.7 - 216.4)	2.0	DOWN
Slc6a8	50690	SLC6A8_32561	141.9 (48.1 - 1438.7)	1.5	DOWN
Meg3	500717	RGD1566401_32721	146.1 (40.1 - 395.1)	1.9	UP
Cenpf	257649	CENPF_8287	169.3 (51.1 - 1294.8)	1.6	DOWN
Cenpw	689399	CENPW_32846	177.3 (58.2 - 542.9)	1.7	DOWN
Ier3	294235	IER3_8864	182.4 (51.2 - 426.7)	1.6	UP
Papss2	294103	PAPSS2_33237	198.6 (83.7 - NC)	1.4	DOWN
Ccl12	287562	CCL12_8217	217.6 (62.0 - 475.4)	1.9	UP

rd25 = 25% relative deviation
NC = Non-convergent

Annotations:

Descriptions of orthologous human genes are shown due to the increased detail that is available in public resources such as UniprotKB (http://www.uniprot.org/uniprot/) and Entrez Gene (https://www.ncbi.nlm.nih.gov/gene/). Human UniprotKB was used as the primary resource due to the greater breadth of annotation and depth of functional detail that is provided. Rat UniprotKB was used as the second resource if the primary source did not provide a detailed description of function. Human Entrez gene summary was used as third resource. Rat Entrez gene summary was used as the fourth resource.
Gene definition version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

Ddit4: HumanUniprotfunction(HumanDDIT4): Regulates cell growth, proliferation and survival via inhibition of the activity of the mammalian target of rapamycin complex 1 (mTORC1). Inhibition of mTORC1 is mediated by a pathway that involves DDIT4/REDD1, AKT1, the TSC1-TSC2 complex and the GTPase RHEB. Plays an important role in responses to cellular energy levels and cellular stress, including responses to hypoxia and DNA damage. Regulates p53/TP53-mediated apoptosis in response to DNA damage via its effect on mTORC1 activity. Its role in the response to hypoxia depends on the cell type; it mediates mTORC1 inhibition in fibroblasts and thymocytes, but not in hepatocytes (By similarity). Required for mTORC1-mediated defense against viral protein synthesis and virus replication (By similarity). Inhibits neuronal differentiation and neurite outgrowth mediated by NGF via its effect on mTORC1 activity. Required for normal neuron migration during embryonic brain development. Plays a role in neuronal cell death. \{ECO:0000250, ECO:0000269;PubMed:15545625, ECO:0000269;PubMed:15632201, ECO:0000269;PubMed:15988001, ECO:0000269;PubMed:17005863, ECO:0000269;PubMed:17379067, ECO:0000269Í PubMed:19557001, ECO:0000269;PubMed:20166753, ECO:0000269;PubMed:21460850\}.

Junb: Human Uniprot function (Human JUNB): Transcription factor involved in regulating gene activity following the primary growth factor response. Binds to the DNA sequence 5'-TGA[CG]TCA-3'.

Cebpd: Human Uniprot function (Human CEBPD): Transcription activator that recognizes two different DNA motifs: the CCAAT homology common to many promoters and the enhanced core homology common to many enhancers (PubMed:16397300). Important transcription factor regulating the expression of genes involved in immune and inflammatory responses (PubMed:1741402, PubMed:16397300). Transcriptional activator that enhances IL6 transcription alone and as heterodimer with CEBPB (PubMed:1741402). \{ECO:0000269;PubMed:1741402\}.

Slc6a8: Human Uniprot function (Human SLC6A8): Required for the uptake of creatine in muscles and brain.

Top 10 Genes Ranked by Potency of Perturbation (Sorted by BMD Median)

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley® SD®

Test Compound: 1,4-Dichlorobenzene (p-dichlorobenzene)
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Lung

Date: 22 May 2025
Time: 12:24:09 PM

Meg3: No description available.

Cenpf: Human Uniprot function (Human CENPF): Required for kinetochore function and chromosome segregation in mitosis. Required for kinetochore localization of dynein, LIS1, NDE1 and NDEL1. Regulates recycling of the plasma membrane by acting as a link between recycling vesicles and the microtubule network though its association with STX4 and SNAP25. Acts as a potential inhibitor of pocket protein-mediated cellular processes during development by regulating the activity of RB proteins during cell division and proliferation. May play a regulatory or permissive role in the normal embryonic cardiomyocyte cell cycle and in promoting continued mitosis in transformed, abnormally dividing neonatal cardiomyocytes. Interaction with RB directs embryonic stem cells toward a cardiac lineage. Involved in the regulation of DNA synthesis and hence cell cycle progression, via its C-terminus. Has a potential role regulating skeletal myogenesis and in cell differentiation in embryogenesis. Involved in dendritic cell regulation of T-cell immunity against chlamydia. \{ECO:0000269;PubMed:12974617, ECO:0000269;PubMed:17600710, ECO:0000269;PubMed:7542657,ECO:0000269;PubMed:7651420\}.

Cenpw: Human Uniprot function (Human CENPW): Component of the CENPA-NAC (nucleosome-associated) complex, a complex that plays a central role in assembly of kinetochore proteins, mitotic progression and chromosome segregation (By similarity). The CENPA-NAC complex recruits the CENPA-CAD (nucleosome distal) complex and may be involved in incorporation of newly synthesized CENPA into centromeres (By similarity). Part of a nucleosome-associated complex that binds specifically to histone H3-containing nucleosomes at the centromere, as opposed to nucleosomes containing CENPA. Component of the heterotetrameric CENP-T-W-S-X complex that binds and supercoils DNA, and plays an important role in kinetochore assembly. CENPW has a fundamental role in kinetochore assembly and function. It is one of the inner kinetochore proteins, with most further proteins binding downstream. Required for normal chromosome organization and normal progress through mitosis. \{ECO:0000250, ECO:0000269; PubMed:19070575, ECO:0000269;PubMed:19533040, ECO:0000269;PubMed:21695110, ECO:0000269;PubMed:22002061,ECO:0000269;PubMed:22304917\}.

Ier3: Human Uniprot function (Human IER3): May play a role in the ERK signaling pathway by inhibiting the dephosphorylation of ERK by phosphatase PP2A-PPP2R5C holoenzyme. Acts also as an ERK downstream effector mediating survival. As a member of the NUPR1/RELB/IER3 survival pathway, may provide pancreatic ductal adenocarcinoma with remarkable resistance to cell stress, such as starvation or gemcitabine treatment. \{ECO:0000269;PubMed:12356731, ECO:0000269;PubMed:16456541, ECO:0000269;PubMed:22565310\}.

Papss2: Human Uniprot function (Human PAPSS2): Bifunctional enzyme with both ATP sulfurylase and APS kinase activity, which mediates two steps in the sulfate activation pathway. The first step is the transfer of a sulfate group to ATP to yield adenosine 5'-phosphosulfate (APS), and the second step is the transfer of a phosphate group from ATP to APS yielding 3'-phosphoadenylylsulfate (PAPS: activated sulfate donor used by sulfotransferase). In mammals, PAPS is the sole source of sulfate; APS appears to be only an intermediate in the sulfate-activation pathway. May have an important role in skeletogenesis during postnatal growth (By similarity). {ECO:0000250}.

Ccl12: Human Uniprot function (Human CCL13): Chemotactic factor that attracts monocytes, lymphocytes, basophils and eosinophils, but not neutrophils. Signals through CCR2B and CCR3 receptors. Plays a role in the accumulation of leukocytes at both sides of allergic and non-allergic inflammation. May be involved in the recruitment of monocytes into the arterial wall during the disease process of atherosclerosis. May play a role in the monocyte attraction in tissues chronically exposed to exogenous pathogens.

** END OF REPORT **